

Chapter 2 First-order DEs

In this chapter we will introduce various techniques to solve some 1st order DEs.

2.2 Separable Equations

A separable DE is an equation that can be written in the form :

$$\frac{dy}{dx} = f(x)g(y) \quad .$$

Note that the right-hand side of the DE can be separated into a function of x times a function of y .

The separable DE $\frac{dy}{dx} = f(x)g(y)$

Can be solved by rewriting it as:

$$\frac{1}{g(y)} dy = f(x) dx$$

and integrating both sides.

DE solution

$$\int \frac{1}{g(y)} dy = \int f(x) dx$$

If we have an IVP with an initial condition $y(x_0) = y_0$, substitute

$$x = x_0 \text{ and } y = y_0$$

into the DE solution to solve for the constant.

ex: ① solve $\frac{dy}{dx} = \frac{x^2}{1-2y}$.

Separate variables and integrate.

$$\int (1-2y) dy = \int x^2 dx$$

$$\Rightarrow \boxed{y - y^2 = \frac{x^3}{3} + C}$$

General Solution.

Note: only one constant of integration is needed.

The solution obtained is an implicit solution.

ex: (2) Solve the IVP $\begin{cases} \frac{dy}{dx} = \frac{2x + 8e^{2x}}{6y^2 + 1} \\ y(0) = 1 \end{cases}$

First, let's solve the DE.

It is a separable equation.

$$\int (6y^2 + 1) dy = \int (2x + 8e^{2x}) dx$$

$$\Rightarrow 2y^3 + y = x^2 + 4e^{2x} + C$$

Let's find C by setting: $x=0$ and $y=1$.

$$\Rightarrow 2 + 1 = 0 + 4 + C \Rightarrow C = -1$$

Then, the solution of the IVP is:

$$\boxed{2y^3 + y = x^2 + 4e^{2x} - 1}$$

ex: ③ solve $\frac{dy}{dx} = \frac{x(1+y^2)}{e^x}$.

It is a separable DE.

$$\int \frac{1}{1+y^2} dy = \int x e^{-x} dx$$

D	I
x	$\oplus e^{-x}$
1	$\ominus -e^{-x}$
0	e^{-x}

$$\Rightarrow \underbrace{\arctan(y) = -xe^{-x} - e^{-x} + C}$$

Implicit solution.

We can solve for y to get an explicit solution.

$$\Rightarrow \boxed{y = \tan(-xe^{-x} - e^{-x} + C)}$$

ex: ④ Solve $\frac{dy}{dx} = 2xy$.

Find an explicit solution.

$$\int \frac{1}{y} dy = \int 2x dx$$

$$\Rightarrow \ln |y| = x^2 + C_1$$

$$\Rightarrow |y| = e^{x^2 + C_1}$$

$$\Rightarrow |y| = e^{C_1} e^{x^2}$$

$$\Rightarrow y = \pm e^{C_1} e^{x^2}$$

↓
relabel as C

$$\Rightarrow \boxed{y = C e^{x^2}}$$

ex: ⑤ Solve the IVP

$$\begin{cases} \frac{dP}{dt} = P^2 - 4P \\ P(0) = 1. \end{cases}$$

Solve for P explicitly as a function of t .

The DE is separable.

$$\underbrace{\int \frac{1}{P^2 - 4P} dP}_{\text{Requires P.F.}} = \int dt$$

Requires P.F.

$$\frac{1}{P^2 - 4P} = \frac{1}{P(P-4)} = \frac{A}{P} + \frac{B}{P-4}$$

$$\text{solve to get: } \begin{cases} A = -1/4 \\ B = 1/4. \end{cases}$$

$$\Rightarrow \int \left[\frac{-1/4}{p} + \frac{1/4}{p-4} \right] dp = \int dt$$

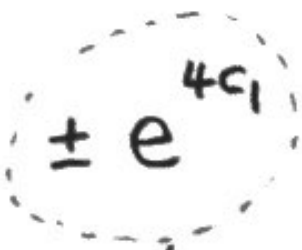
$$\Rightarrow -\frac{1}{4} \ln|p| + \frac{1}{4} \ln|p-4| = t + C_1$$

$$\Rightarrow -\ln|p| + \ln|p-4| = 4t + 4C_1$$

$$\Rightarrow \ln \left| \frac{p-4}{p} \right| = 4t + 4C_1$$

$$\Rightarrow \left| \frac{p-4}{p} \right| = e^{4C_1} e^{4t}$$

$$\Rightarrow \frac{p-4}{p} = \pm e^{4C_1} e^{4t}$$



 Relabel as C

$$\Rightarrow \frac{p-4}{p} = C e^{4t}$$

Let's find C from $P(0) = 1$.

set $t = 0$ and $P = 1$.

$$\Rightarrow \frac{1-4}{1} = C e^0 \Rightarrow C = -3$$

$$\Rightarrow \frac{P-4}{P} = -3 e^{4t}$$

Solve for P

$$\Rightarrow P - 4 = -3 P e^{4t}$$

$$\Rightarrow P + 3 P e^{4t} = 4$$

$$\Rightarrow P(1 + 3 e^{4t}) = 4$$

$$\Rightarrow \boxed{P = \frac{4}{1 + 3 e^{4t}}}$$